Real World Scenarios with HANA Cloud

Meet the Speaker

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BDC Blackbelt MEE

How **old** is SAP HANA?

How **old** is SAP HANA?

How **long** are you part of the HANA story.

How old is SAP HANA?

SAP HANA is **approximately 16 years old**.

Timeline

- **2010**
- **2011:** GA
- **2015: SAP S/4HANA**
- **2018: SAP HANA 2.0**,
- **2019–2021: SAP HANA Cloud**
- **Today (2026):** SAP HANA is SAP's strategic database platform, powering SAP S/4HANA and a wide range of analytics, data integration, AI, and cloud-native applications.

Despite being introduced more than 16 years ago, SAP HANA continues to evolve, with most new innovations focused on **SAP HANA Cloud** while many customers continue to run **SAP HANA 2.0** on-premises.

Which are the Use Cases for HANA?

Run:

ECC, S4, BW, SCM, DSP, SAC, IDB, ….

Which are the Use Cases for HANA?

Run:

ECC, S4, BW, SCM, DSP, SAC, IDB, …:

But it is also the „best“ (Database) Platform → So our customers are using it also for

Custom Use Cases ->
Let‘s take a look what they are doing!

What are the Problem Statements?

How can I guide you?

Transitioning from expensive on-premise HANA sidecars to scalable, cost-effective cloud data environments.

Migrating complex HANA IP (within calc views) into a self-service model without losing critical business logic.

Applying Generative AI to "Clean Core" architectures to drive cross-functional usability and efficiency.

Empowering field teams to generate instant, intelligent sales quotes using real-time customer and credit data.

Building intelligent ChatBots that unify SAP ERP data with unstructured contracts and PDFs to accelerate offers.

Building transactional side car solutions with data context from SAP and non-SAP sources

SAP HANA Cloud

Scenarios

01 EDWH, ODS or Data Platform

(Big) Data Scenarios in the Cloud

- Seamless lift-and-shift
- Preserve SAP investments
- Self-service migration tools

02 Apps & Agents

Application & Agent Development

- AI-infused apps & agents
- Live transactional data
- Multi-model capabilities

03 ML, KG, VE, Advanced Analytics

Advanced SQL Analytics & ML

- In-database AI/ML (PAL/APL)
- Python & SQL data science
- Governed BDC data products

Multi-Model, AI/ML, and Live Business Data - All in one Database in the Cloud.

Migration of Database Artefacts

Migration of Database Objects

Compatibility Analysis

Type	PRD	
# of Delivery Units	36	NA
# of Users		
# of Roles	700	
# of Schemas	200	
Runtime Objects		
Tables	17,120	
Partitions	13,000+	
Procedures	7	
Analytic Privileges	TBD	
Attribute Views	-	
Analytics Views	-	
Calculation Views	6,000+	
Virtual Tables	3,083	

High level compatibility reference :
https://help.sap.com/docs/hana-cloud/sap-hana-cloud-migration-guide/high-level-feature-compatibility

Type	PRD	
Design Time Objects		
hdbtable	176	
hdbprocedure	191	
hdbrole	143	
hdbscalarfunction	-	
hdbschema	13	
hdbsequence	33	
hdbstructure		
hdbtablefunction	83	
hdbti	93	
hdbflowgraph	3	
hdbview	153	
project	-	
xsaccess	-	
xsapp	-	
xshttpdest	4	
xsjs	204	
xsjslib	540	
xsodata	63	
xsprivileges	37	
xssecurestore	-	
xssqlcc	-	

Direct migration possible

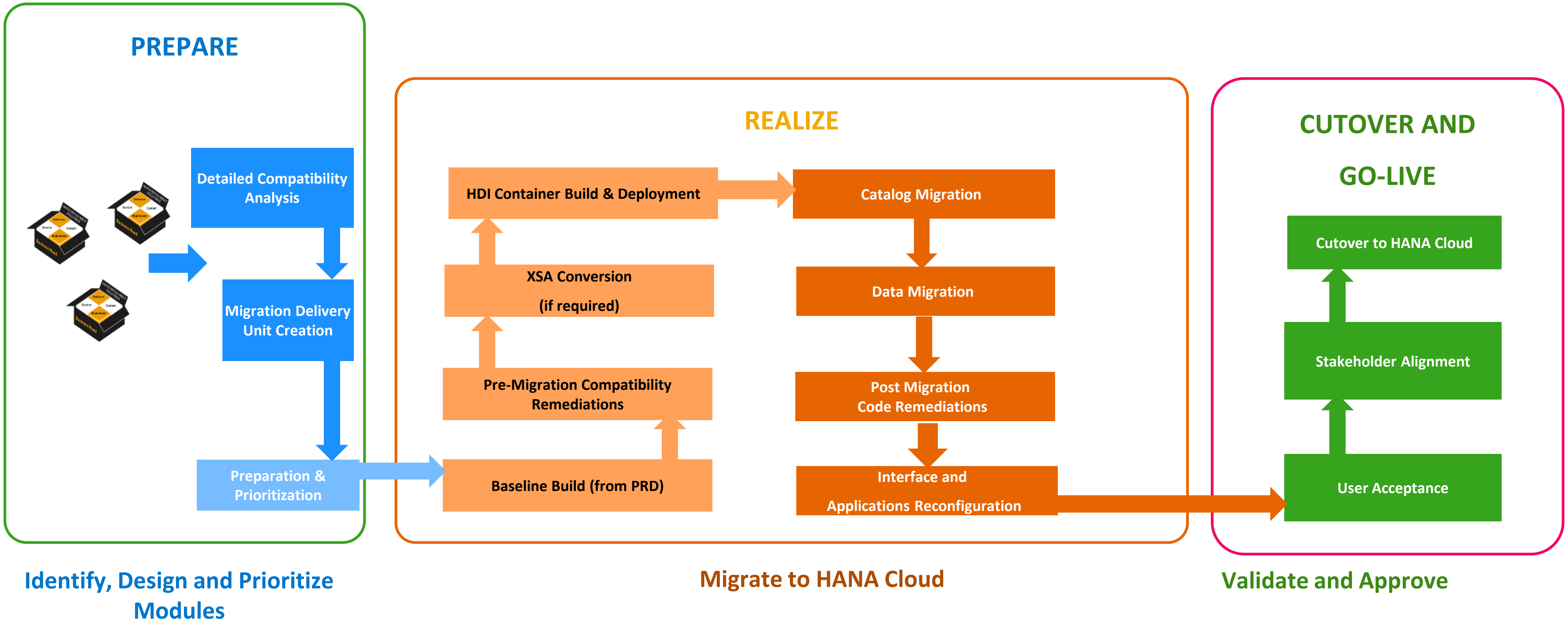
Corrections may be required for HC compatibility

Type not supported directly in HC, need conversion to a supported type

Not Applicable for HC

Tool support available

Recommended Approach for HANA Cloud Migration

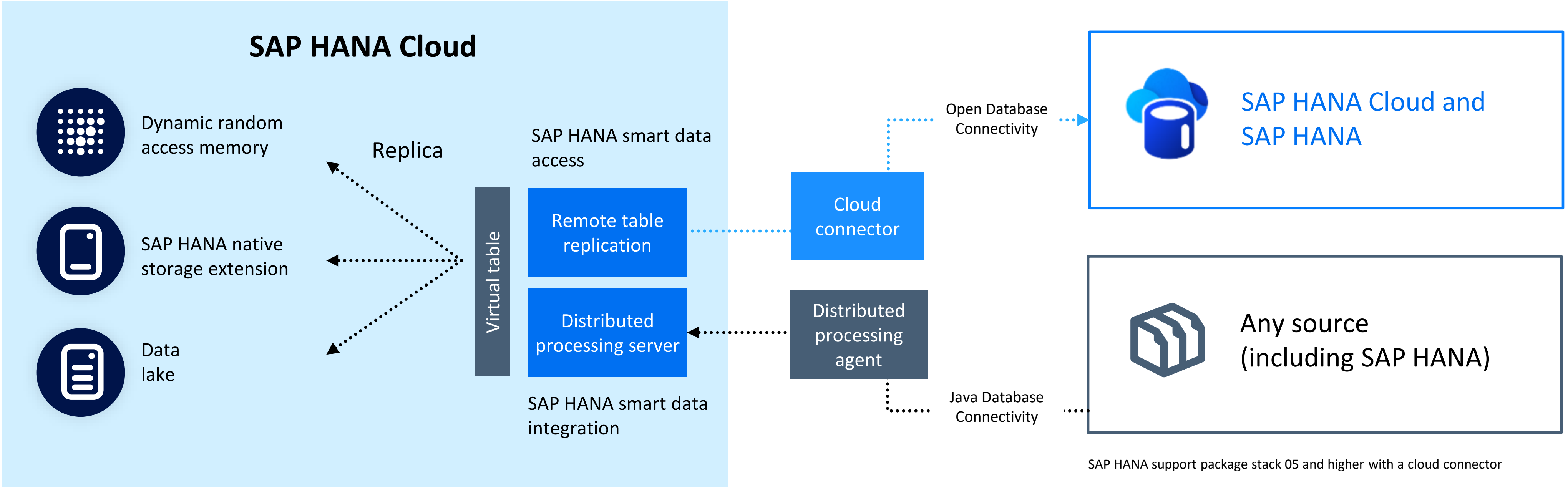


WebIDE ➡ BAS

How can data be integrated

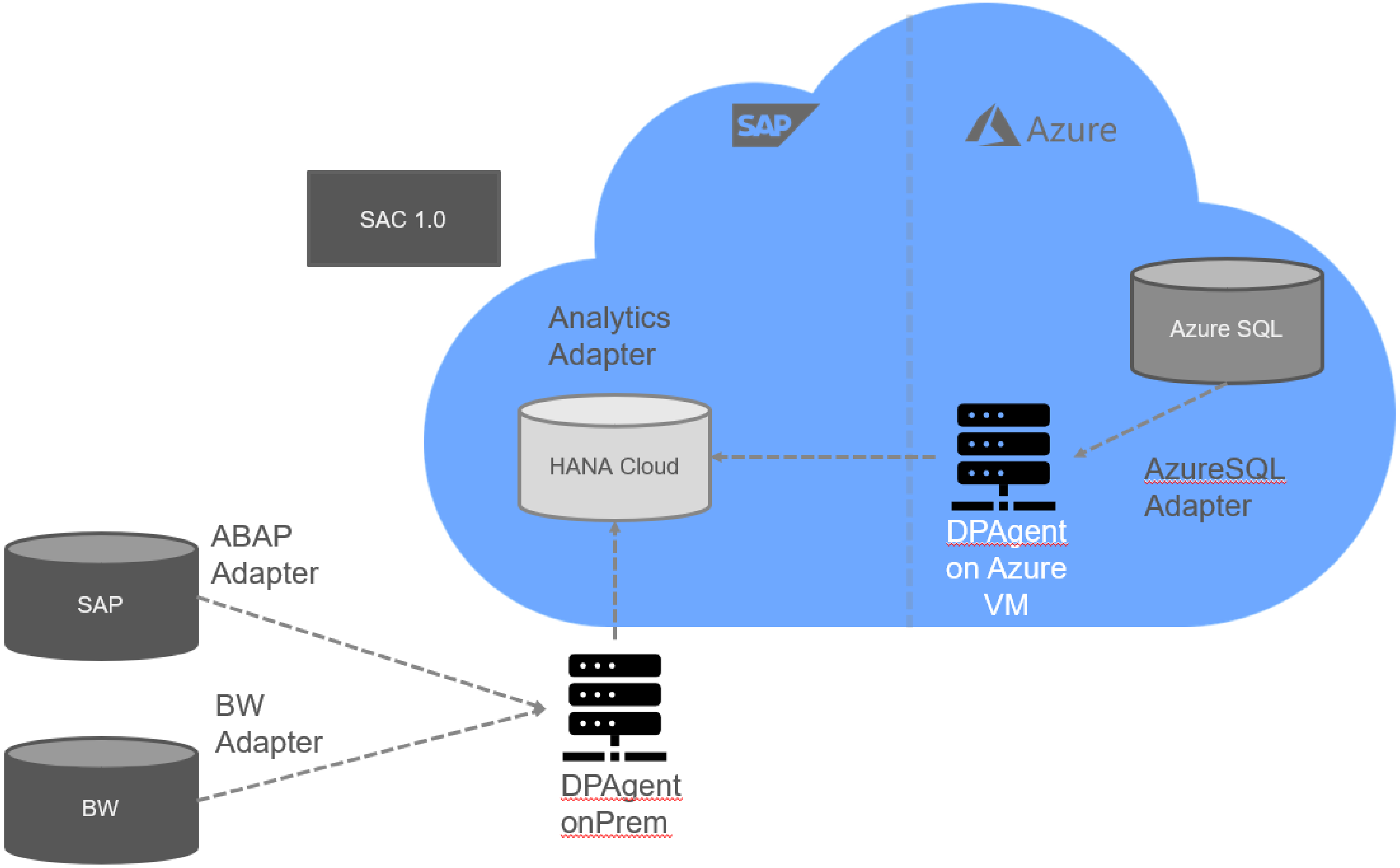
SAP HANA Cloud | Hybrid landscapes

Optimized and enhanced replication



- A common SQL interface enables remote table replication with SAP HANA smart data access and standard replication based on SAP HANA smart data integration
- SAP-HANA-to-SAP-HANA replication is optimized with remote table replication (native engine feature of SAP HANA) using transaction layer logs
- Flexible connectivity options are available via standard SAP HANA smart data integration replication, supporting various sources
- Flexible storage options include table replicas in SAP HANA Cloud (in-memory or native storage extensions)
- Remote table replication is supported between SAP HANA support package stack 05 and higher and SAP HANA Cloud (through a cloud connector)

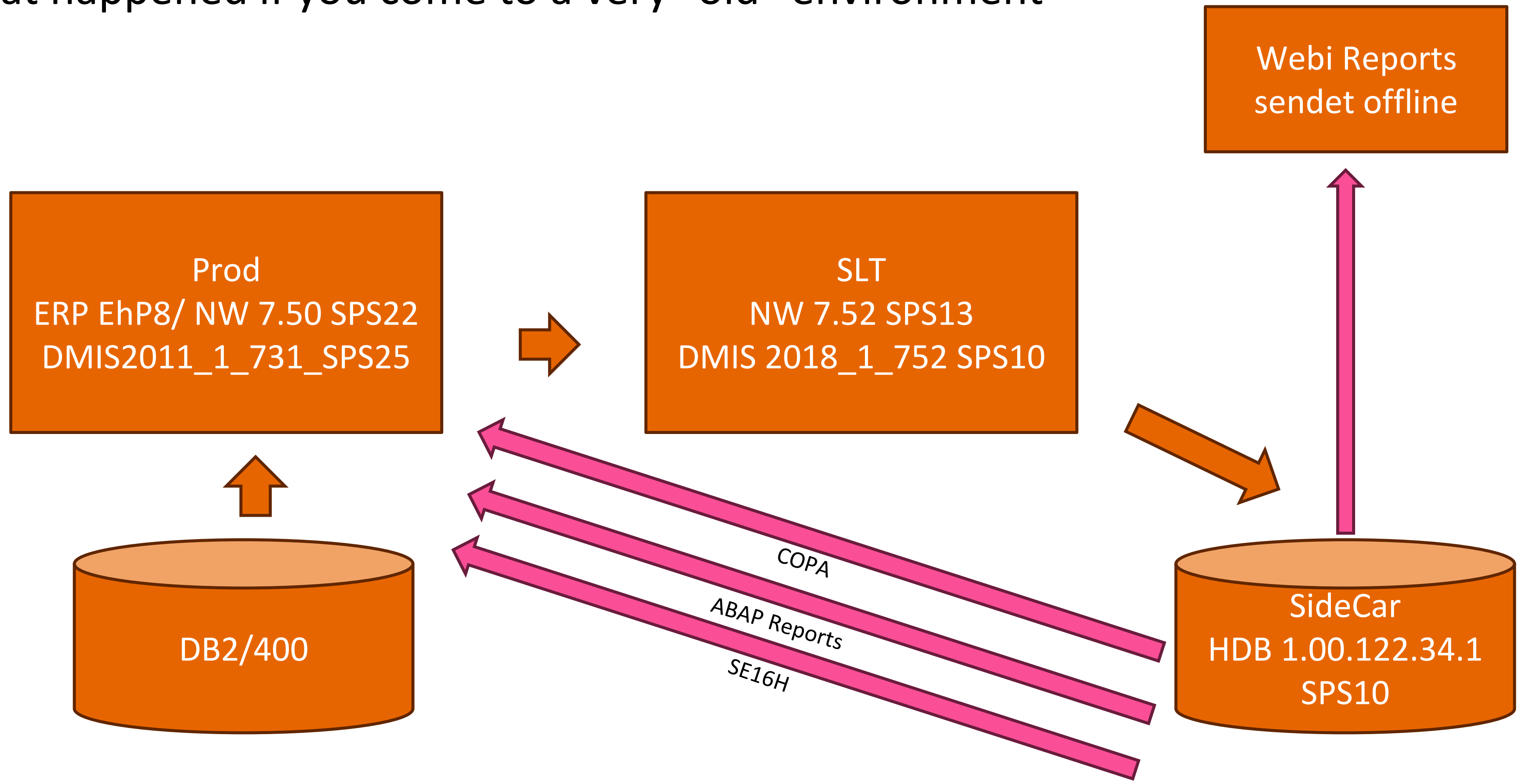
HANA Cloud Szenario – Architecture



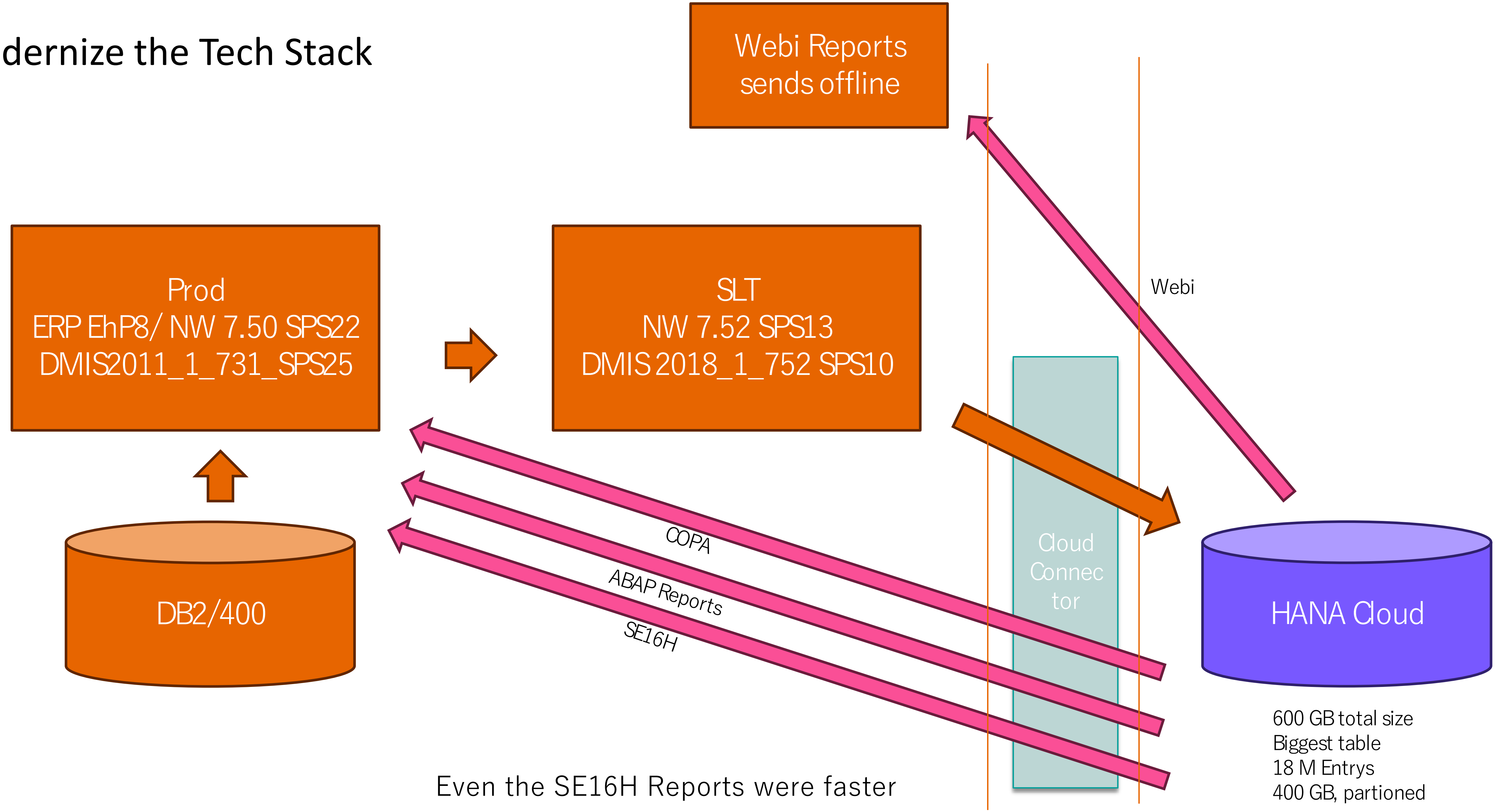
Findings from a very big PoC -> Real life

1. HANA Cloud and SDI have consistently prooven not to be the performance bottleneck.
2. A single SDI replication connection was able to fully utilize the customer's available outbound internet bandwidth.
3. Even multi-terabyte data volumes are not a concern. We have successfully transferred data at rates of up to **4 TB per hour**, fully saturating the customer's outbound network connection.
4. SDI not only replicating data – also fast and reliable file upload is supported
5. No need for a DMZ

What happened if you come to a very “old“ environment



Modernize the Tech Stack



Even the SE16H Reports were faster

Than before!

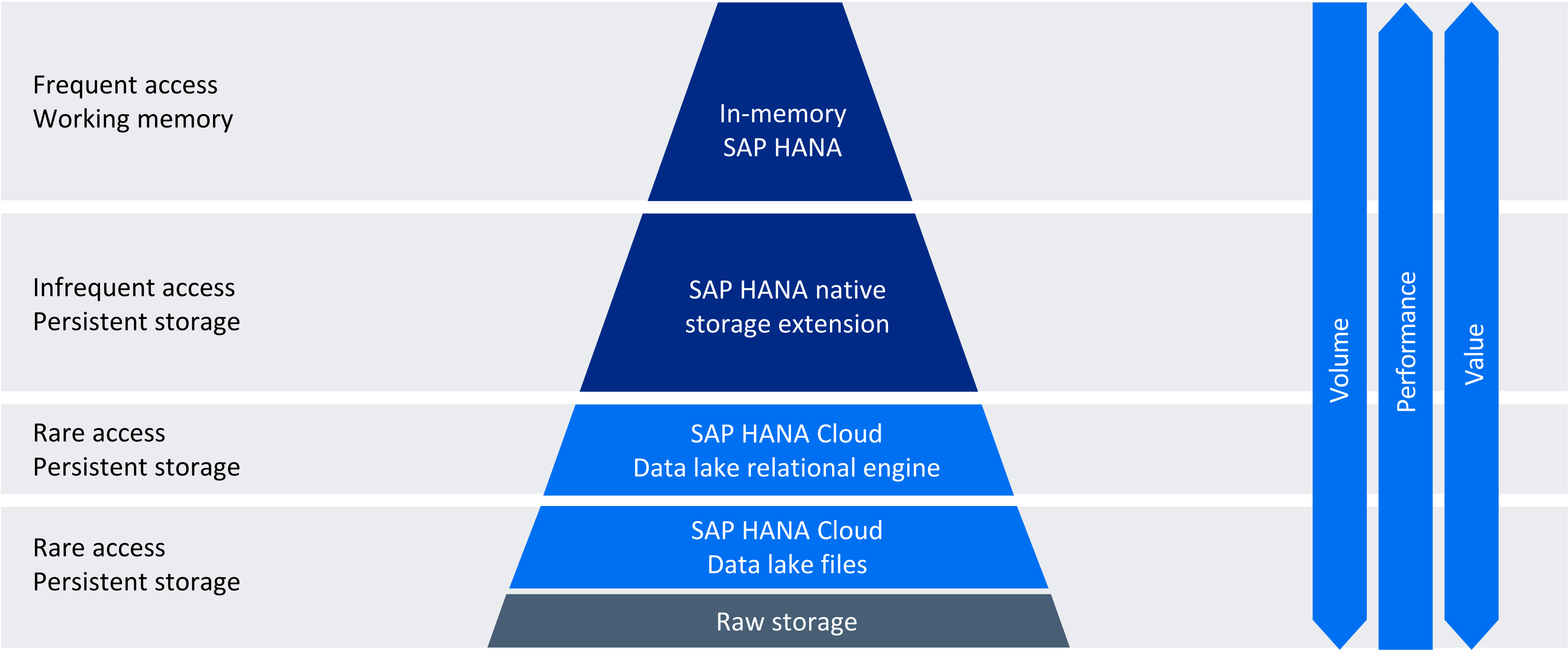
Findings

- HANA Cloud can be accessed from all „open“ SAP Application (ECC, SLT, S4, ··
- HANA Cloud can access every HANA oP Instance
- HANA Cloud is compliant to access every HANA oP Instance
- And -> Its fast and very easy to integrate data

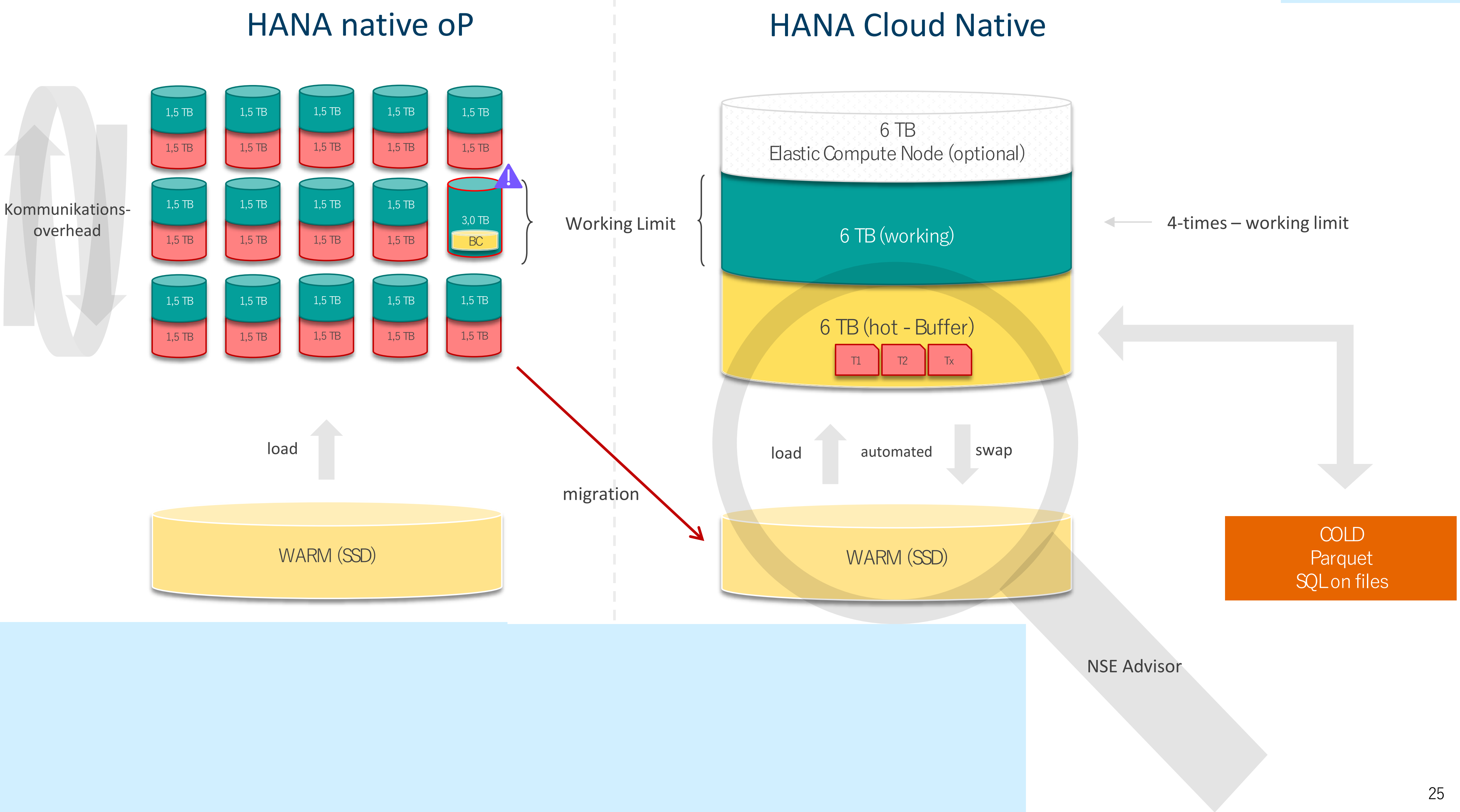
HANA Cloud Instances are too small

SAP HANA Cloud | Data pyramid

Data storage for various use cases



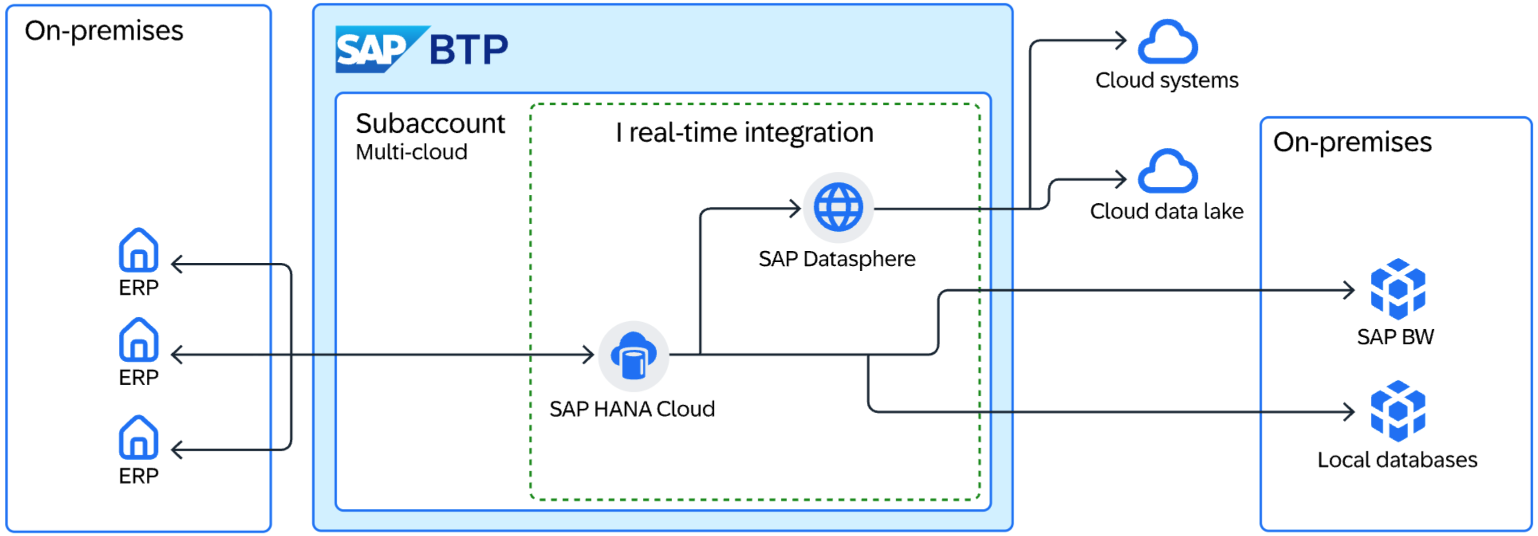
Comparison of Architecture



Manufacturing

Real-Time Data Integration as a Foundation for Scalable Insights

A global industrial manufacturing company transformed its data landscape to enable real-time insights across the enterprise. Facing increasing complexity from fragmented systems and growing data volumes, the organization implemented a modern data integration architecture. This shift unlocked faster decision-making, improved operational efficiency, and created a scalable foundation for future innovation.



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SAP HANA Cloud

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- Self-service migration tools

02 A p p s & A g e n t s

Application & Agent Development

- AI-infused apps & agents
- Live transactional data
- Multi-model capabilities

03 M L , K G , V E , A d v a n c e d A n a l y t i c s

Advanced SQL Analytics & ML

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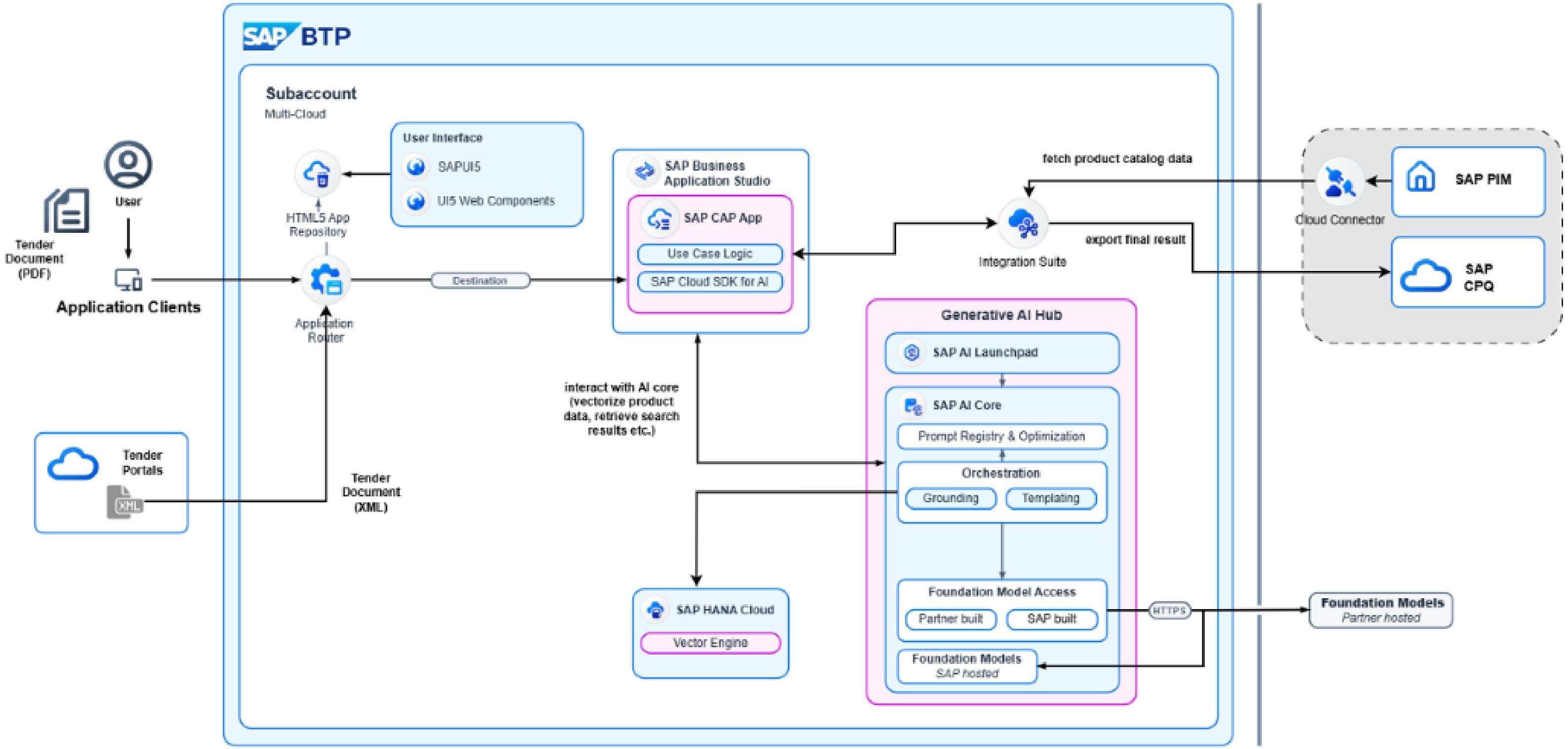
Multi-Model, AI/ML, and Live Business Data - All in one Database in the Cloud.

SAP HANA Cloud

Large Enterprise

AI-Powered Invoice Verification to Prevent Duplicate Payments

A large enterprise transformed its accounts payable process by implementing an AI-driven solution to detect duplicate invoices. By vectorizing every invoice and comparing new submissions against existing ones, the company ensures that potential duplicates are identified before payment, improving financial control and efficiency.



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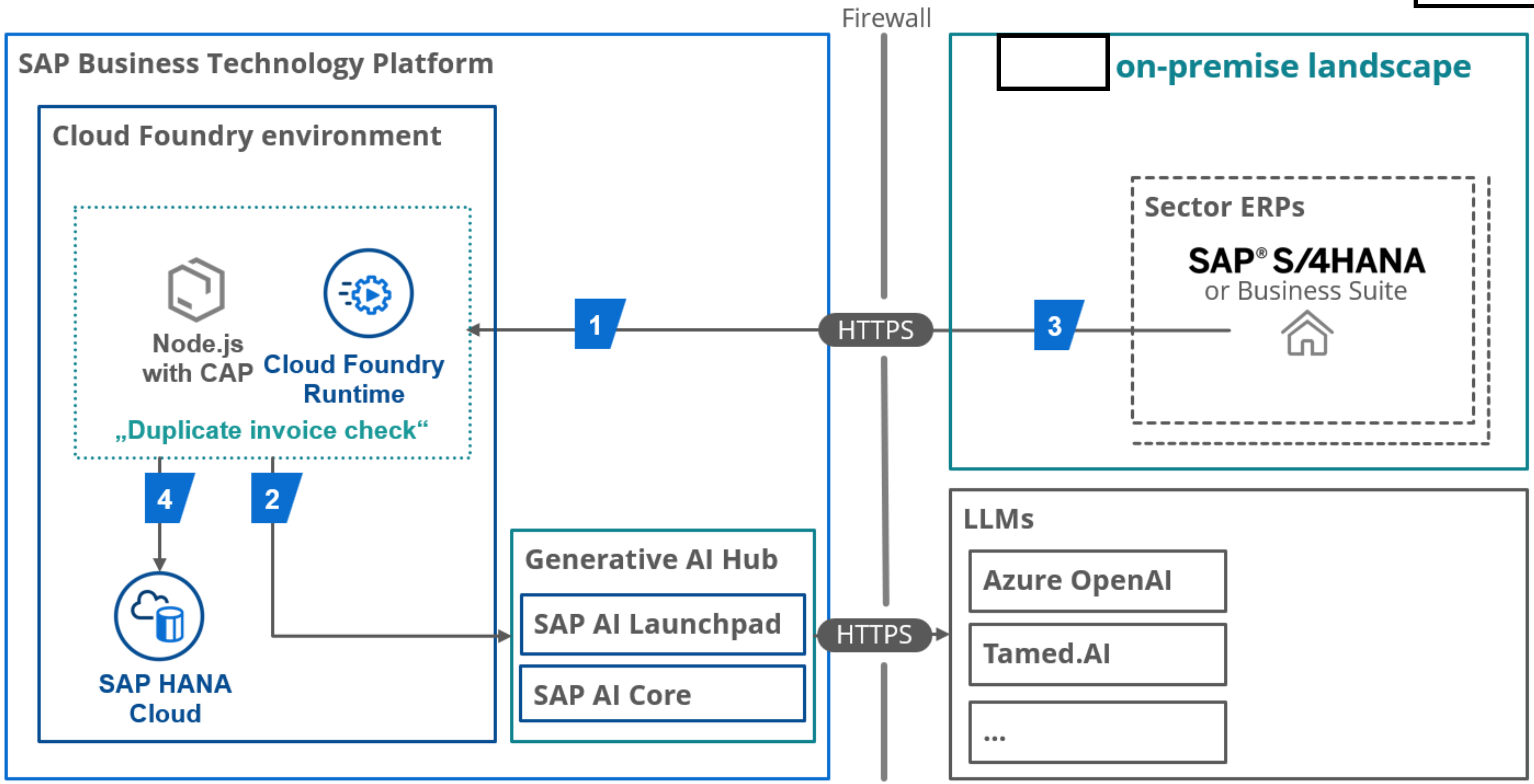
USE CASE: PREVENTING DUPLICATE PAYMENT

Use case steps

1. Invoice posting send via open FI-Event 1030 to duplicate check
2. Check against stored invoices with AI logic
3. Depending on probability setting payment block
4. Store posted document to HANA Cloud



For simplification of the diagram further required service like connectivity, destination, event mesh, identity provider aren't incorporated.

SAP HANA Cloud

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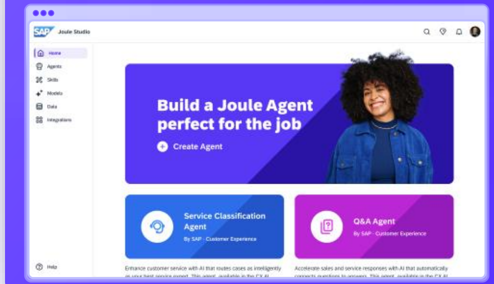
AI Foundation for use in custom AI scenarios

Leverage a powerful set products to build, connect, and run custom AI solutions and agents.

Build agents and extend Joule.

Facilitate smooth, low-code customizations.

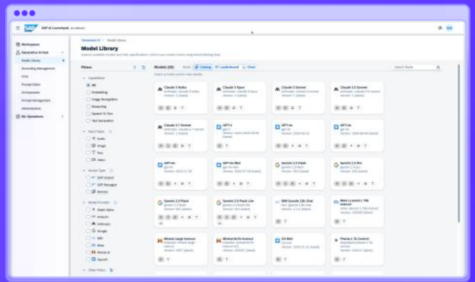
Joule Studio in SAP Build



Explore, orchestrate, and productize AI.

Experiment with models and productize gen AI at scale.

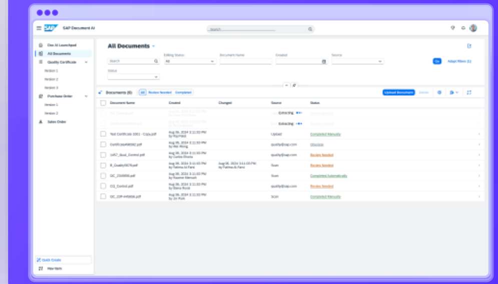
Generative AI hub



Automate document-based processes.

Process structured and unstructured documents.

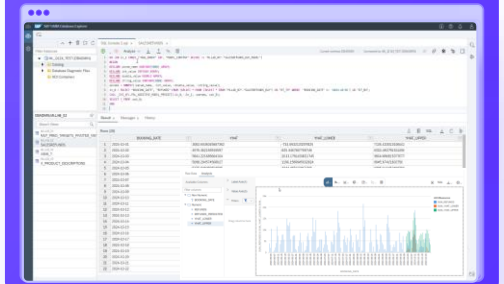
SAP Document AI



Run AI workloads on databases.

Eliminate data movement and processing delays.

SAP HANA Cloud



Core technologies

SAP Knowledge Graph

SAP Foundation Model

and more

Resources and data

SAP and third-party data through SAP Business Data Cloud, SAP HANA Cloud

Infrastructure

Remote services

Cash Flow Forecast

Consumer Products, Switzerland

Benefit Increased insight into future Cash Flow

It was proven, that SAP Business Data Cloud can provide a scalable mid- to long-term cash forecasting available on a weekly and monthly basis. The in-memory performance allows for new spontaneous analysis even on large datasets (150 million records). The various built-in Machine Learning algorithms give flexibility and choice for the implementation.

Solution Custom BTP AI Solution

SAP Business Data Cloud was used as data processing and Machine Learning platform. Forecasts are shared with the Finance department through SAP Analytics Cloud. A planning grid enables the end users to manually adjust the predictions, to account for unusual constellations that cannot be predicted.

Challenge Limited visibility into mid- and long-term future Cash Flow

Increased transparency into future Accounts Receivable and Account Payable amounts can allow for more long term investment decisions, ensuring that sufficient liquidity is available without keeping excessive amounts in unprofitable short term liquidity.

OUT-OF-THE-BOX AI FUNCTIONS FOR TABULAR DATA

Python Machine Learning client for SAP HANA

Native Python machine learning client for SAP HANA Cloud*

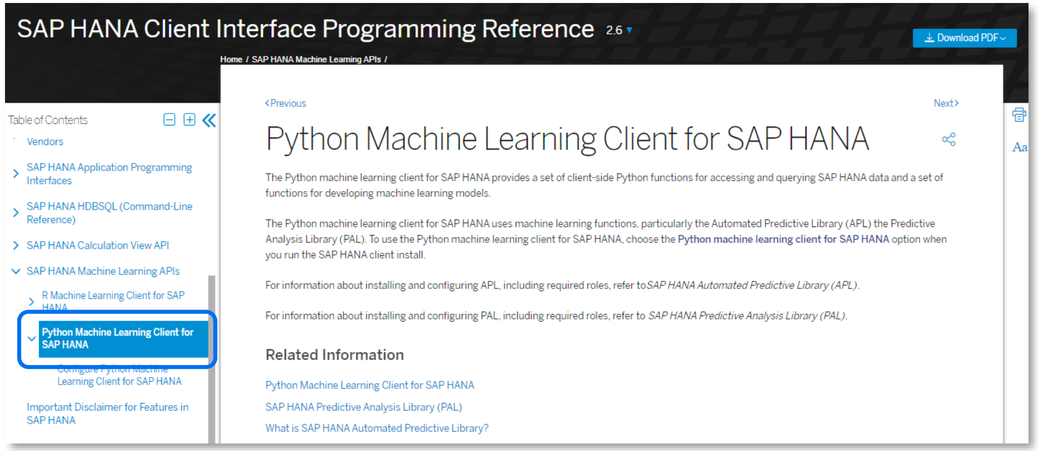
– Allows to script-in-python,
while SQL is generated and executed on the fly

– Expose SAP HANA data as HANA dataframe in Python

– Database AI library functions of Predictive Analysis Library (PAL) and Automated Predictive Library (APL) exposed and Python methods

– Remote use of SAP HANA's machine learning-, spatial- and graph functions in Python

– Available with SAP HANA Client download
+ expanded distribution via PyPi
https://pypi.org/project/hana-ml/



Machine Learning APIs in SAP HANA Client Interface Programming Reference

*Support for SAP HANA Cloud and SAP HANA Platform

Python Machine Learning client for SAP HANA

Classification scenario example

- Leveraging PAL unified classification procedure
  - single interface for Decision trees, Hybrid gradient boosting tree, Logistic regression, Multi-class logistic regression, Naïve Bayes, Random decision trees, Support Vector Machine
  - Includes optimal model parameter selection, fit / score / predict functions, additional debrief statistics and metric data for model value plot, …

Fit model

```
In [ ]:  rdt_params = dict(random_state=2, split_threshold=1e-7,
                          min_samples_leaf=1,  n_estimators=10,
                          max_depth=55)

         uc_rdt = UnifiedClassification(func = 'RandomForest',
                                        **rdt_params)

         uc_rdt.fit(data=diabetes_train,
                    key= 'ID',
                    label='CLASS',
                    partition_method='stratified',
                    stratified_column='CLASS',
                    partition_random_state=2,
                    training_percent=0.7, ntiles=2)
```

Score and evaluate model

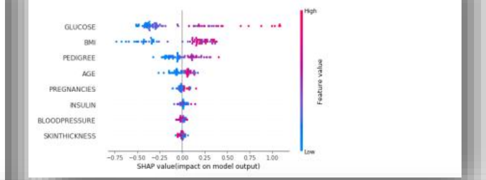
```
_,_,_,metrics_res = uc_rdt.score(data=diabetes_test, key='ID', label='CLASS')
metrics_res.collect()
```

```
uc_rdt.statistics_.collect()
```

	STAT_NAME	STAT_VALUE	CLASS_NAME
0	AUC	0.8089112712665406	None
1	RECALL	0.8571428571428571	0
2	PRECISION	0.7611940298507462	0
3	F1_SCORE	0.8063241106719368	0
4	SUPPORT	119	0
5	RECALL	0.5076923076923077	1
6	PRECISION	0	[illegible]
7	F1_SCORE	0.5739130434782	[illegible]
8	SUPPORT	[illegible]	[illegible]
9	ACCURACY	0.73369565217391	[illegible]
10	KAPPA	0.38499317871759	[illegible]
11	MCC	0.39200305267712	[illegible]

```
from hana_ml.visualizers.model_debriefing import TreeModelDebriefing

shapley_explainer = TreeModelDebriefing.shapley_explainer(pred_res, diabetes_test, key='ID', label='CLASS')
shapley_explainer.summary_plot()
```




```
uc_rdt.metrics_.collect()
```

	NAME	X	Y
0	RANDOM_CUMGAINS	0.0	1.000000
1	RANDOM_CUMGAINS	1.0	1.000000
2	RANDOM_LIFT	0.0	1.000000
3	RANDOM_LIFT	1.0	1.000000
4	RANDOM_CUMLIFT	0.0	1.0000
5	RANDOM_CUMLIFT	1.0	1.0000
6	PERF_CUMGAINS	0.0	0.0000
7	PERF_CUMGAINS	0.5	1.0000
8	PERF_CUMGAINS	1.0	1.0000
9	PERF_LIFT	0.0	2.0000
10	PERF_LIFT	0.5	2.0000
11	PERF_LIFT	1.0	[illegible]

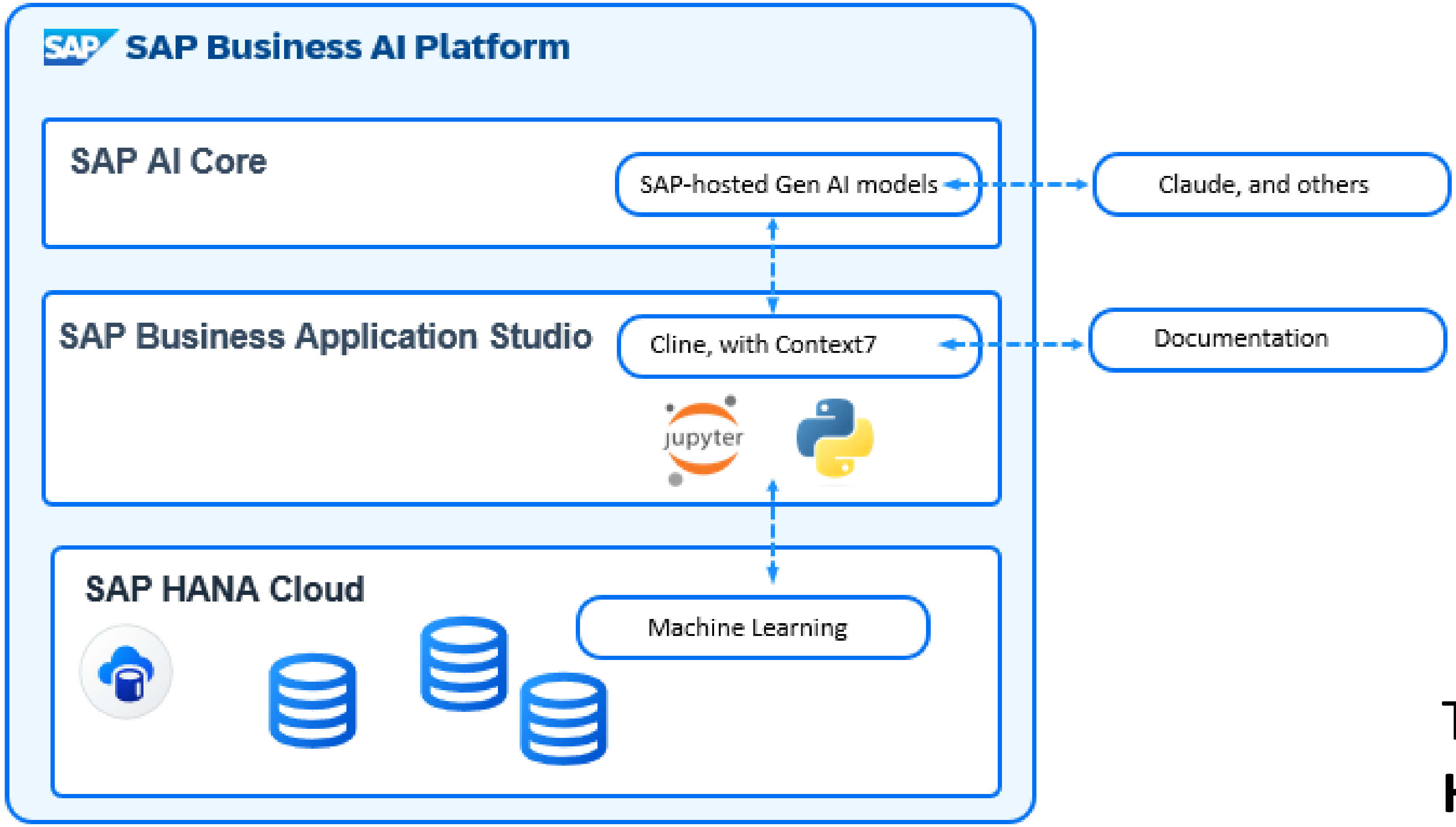
RANDOM_CUMGAINS
RANDOM_LIFT
RANDOM_CUMLIFT
PERF_CUMGAINS
PERF_LIFT
PERF_CUMLIFT
ROC_FPR
ROC_TPR
CUMGAINS
LIFT
CUMLIFT

Predict incl. explainability

```
pred_res = uc_rdt.predict(diabetes_test, key='ID', features=features)
pred_res.head(10).collect()
```

	ID	SCORE	CONFIDENCE	REASON_CODE
0	1	0	0.6	{"GLUCOSE":0.1391(36%),"PREGNANCIES":-0.0890(2...
1	2	0	0.7	{"AGE":-0.0746(17%),"GLUCOSE":0.0715(16%),"BMI...

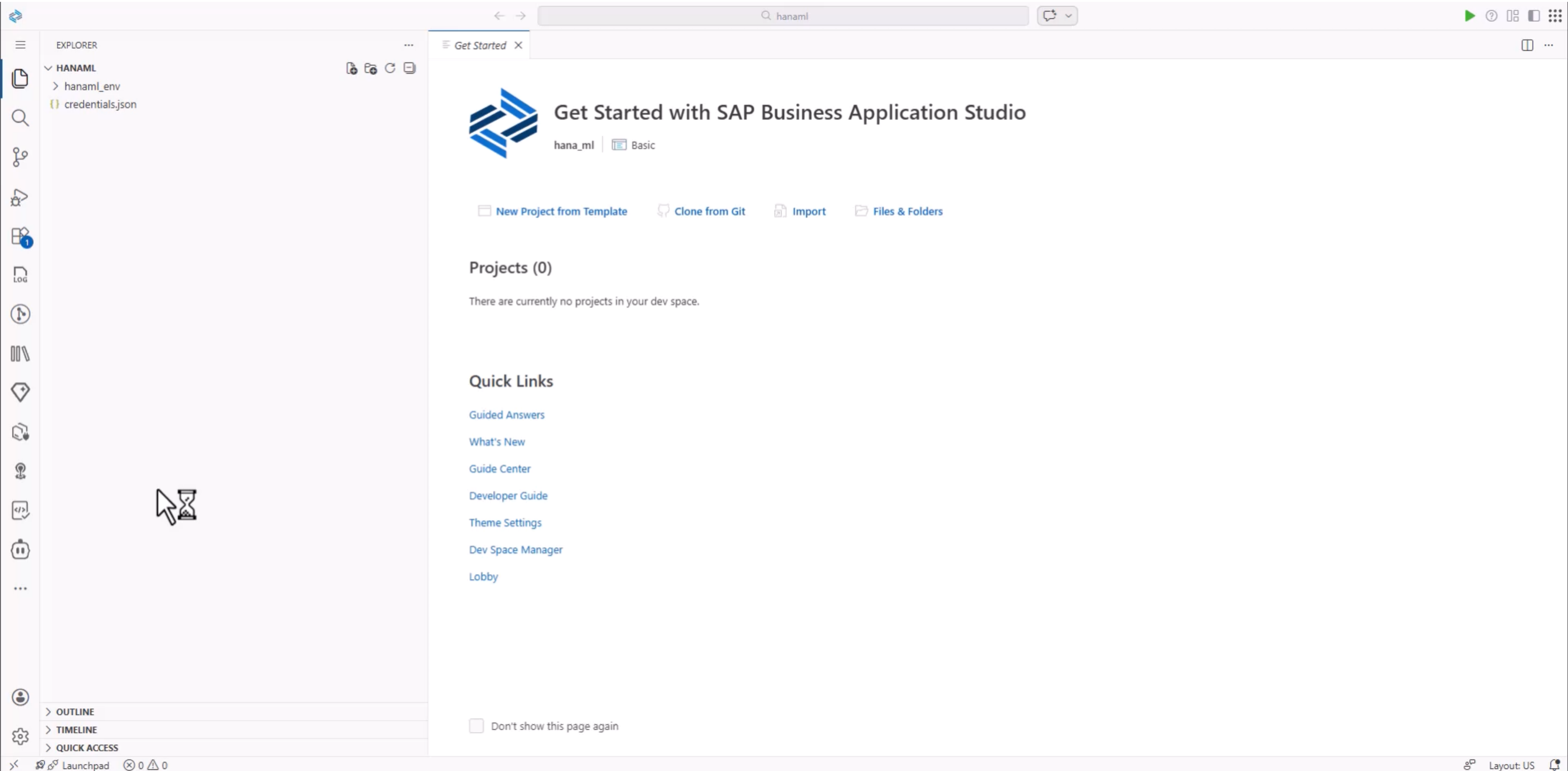
Vibe coding SAP HANA Cloud Machine Learning with SAP Business Application Studio



Try out
Hands-on tutorial

Create a new notebook called "HANA ML Time-series forecast". Use the details from @/credentials.json to and the hana_ml package to connect to SAP HANA Cloud. Connect to the table HOTELNIGHTS. Do not download the data. Use hana_ml to have HANA Cloud aggregate this table on the MONTH column, summarising the column HOTELNIGHTS. Use the hana_ml Package to train an AdditiveModelForecast model and predict the next 6 months. Join the actuals and the prediction in a single structure and save as table HOTELNIGHTS_FORECAST_BAS. Test the code and fix any errors.

Vibe coding: Time-series forecast with HANA ML

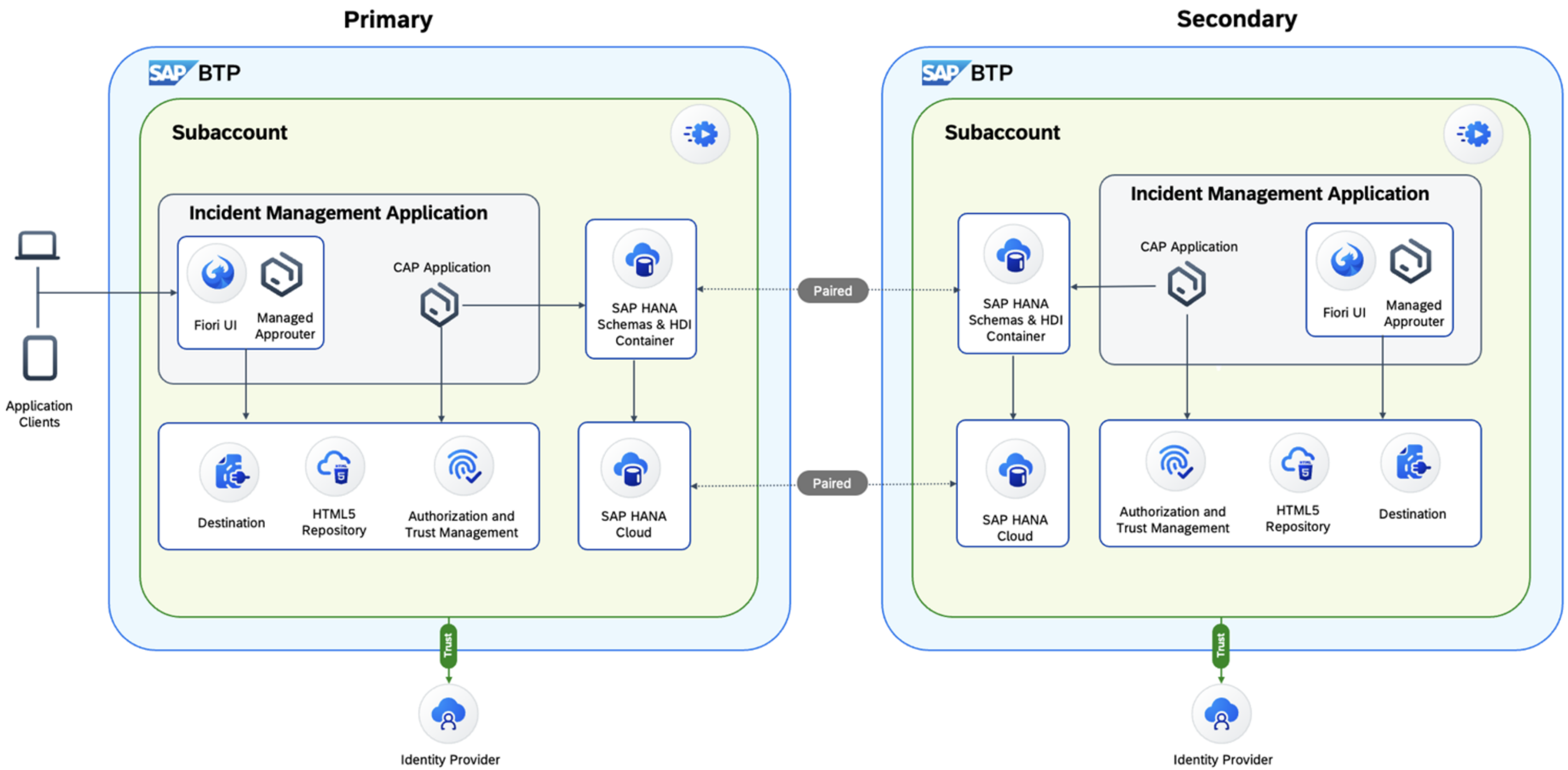


One More Thing …

Mission-Critical Workloads

Multi-Region Business Continuity

A multi region disaster recovery setup ensures business continuity for mission critical applications by running a fully redundant app and database landscape in two separate regions. It protects against outages caused by cyberattacks, hardware failures, or natural disasters, where downtime would have major financial, reputational, and regulatory impact. SAP HANA Cloud replicates data asynchronously to a secondary region, and a global traffic manager enables fast failover and traffic redirection. This delivers significantly higher resiliency with minimized data loss and business continuity at your hands.



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Thank you

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